

Carrier Sales Automation Build Description

Prepared for: Acme Logistics

Overview

This build is a prototype for automating inbound carrier calls related to load booking. The system is designed to verify carriers, retrieve load information, negotiate pricing, and capture operational metrics from every completed conversation.

Use Case

When a carrier calls in, the agent:

1. requests the MC number
2. verifies the carrier through an FMCSA-backed lookup flow
3. requests the broker reference or load identifier
4. retrieves structured load details from the broker load API
5. negotiates pricing within the workflow
6. transfers or closes the call based on the booking outcome
7. writes completed call data into the reporting backend

System Components

- Inbound HappyRobot workflow for the call logic
- Carrier verification API for MC lookup
- Load lookup API backed by a broker-managed load dataset
- Completed-call webhook for analytics ingestion
- PostgreSQL database for metrics persistence
- Live dashboard for reporting and QA review

Deployment

- Cloud provider: AWS
- Runtime: ECS Fargate
- Database: PostgreSQL on RDS
- HTTPS endpoint: <https://api.hrsubs.dev>
- Dashboard: <https://api.hrsubs.dev/dashboard/>

Security

- HTTPS is enabled for deployed traffic
- Business endpoints require API key authentication via X-API-Key
- Secrets are injected through AWS Secrets Manager in deployment

Dashboard Metrics

The dashboard supports custom reporting beyond the workflow itself, including:

- booking rate
- negotiation rounds
- call duration
- final agreed rate
- lane distribution
- recent call history
- sentiment mix

Prototype Scope

For the prototype, load data is served from a structured broker-owned dataset exposed through an API. In production, the same interface could be backed by a TMS, internal database, or another load management service.

Operational Notes

- The completed-call webhook is idempotent by `call_id`
- Workflow retries update the same record rather than creating duplicates
- Dashboard metrics can be reset for clean demo runs

Outcome

This prototype demonstrates a workable inbound carrier sales flow with both operational automation and a reporting layer suitable for internal review, QA, and broker-side visibility.